CLEO: ten campaigns, one trade

Narrative deck — compiled November 4, 2026

Abstract

One panel per experimental campaign, on matched axes where the axes are genuinely matched. The recurring quantity is a Pareto trade: oracle pass rate against sequence diversity, first over single designs and then over the recombined library that the combinatorial filter actually produces. The closing section names three narrative arcs these panels will carry, with the panel that argues against each. This document is for choosing the argument, not for making it.

How to read every panel. Up is a library with fewer dead wells. Right is a library covering more sequence space. U_{15} is the count of distinct (position, residue) substitutions rarefied to 15 passing designs over 300 draws, measured against the per-position modal residue of the LigandMPNN temperature sweep on the same backbone — the reference under which the published baseline comparator reproduces. Library diversity is the same quantity over passing library members, rarefied to a matched count, so an arm cannot win it merely by having more of its library fold.

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1 C1 — Temperature is the dial everyone already has

`experiments/ame/tempsweep · M0097_1ctt · LigandMPNN, 96 designs per T`

The comparator campaign. Sampling temperature is how diversity is bought out of LigandMPNN today, and it charges pass rate for every point of coverage.

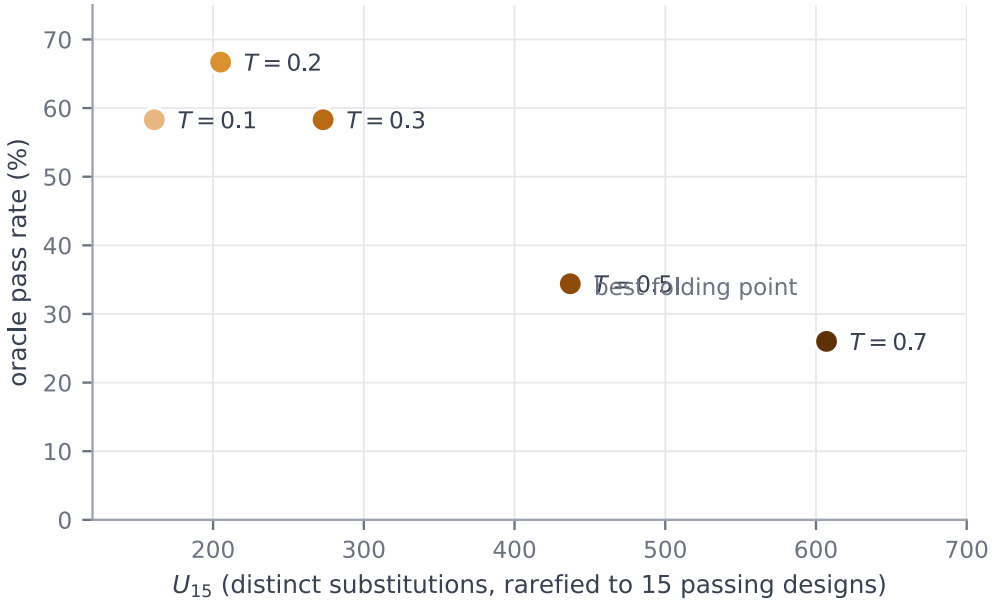


Figure 1: Six independent operating points, shaded light to dark by temperature. They are deliberately not joined: a line between them would assert a continuity nobody measured. $T = 1.0$ is omitted — at 4.2 % pass, too few designs survive to rarefy to 15; its mean pairwise Hamming is 126.3.

What it shows. The best diversity the baseline reaches while still folding is $U_{15} = 607$ at 26.0 % pass, at $T = 0.7$. Past that the curve falls off a cliff.

What it can't say. One backbone. $T = 0.3$ scores *below* $T = 0.2$ on pass rate, so even the baseline curve is not monotone at $n = 96$ — read the shape, not the individual points.

2 C2 — The KL anchor is the dial RL adds

`experiments/ame/klsweep_table.py · run_M0097_1ctt_cond9_14_klw* · 200 steps, 640 designs per arm`

Six arms of one config differing only in `kl_weight`. The anchor holds the policy near the pretrained model, so raising it trades pass rate back for coverage — the same trade as temperature, on a different mechanism.

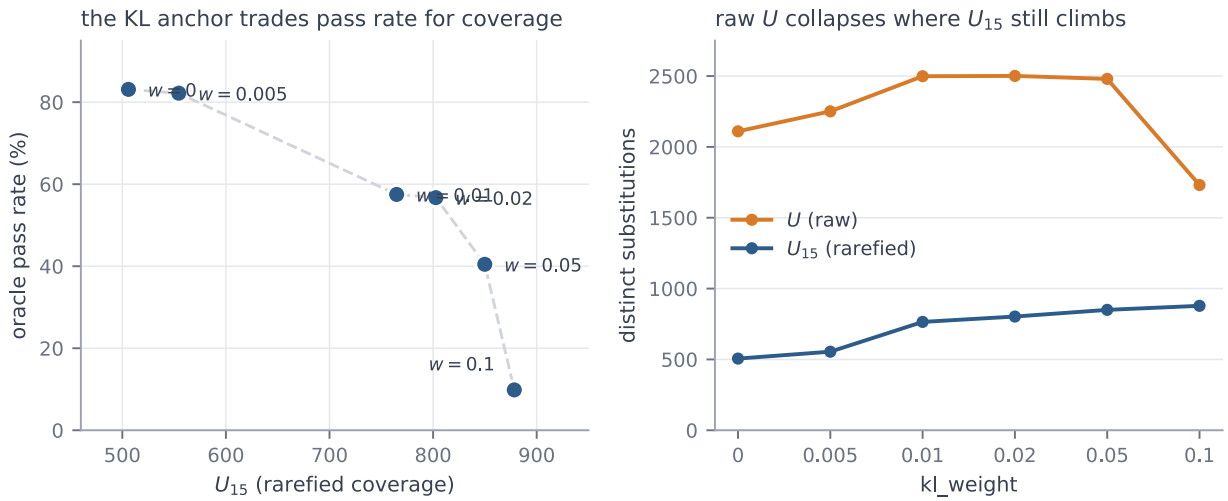


Figure 2: Left: dashed segments show the ordering of `kl_weight` only, not an interpolation. Right: raw U peaks at $w = 0.02$ and then *falls* to 1731 at $w = 0.10$, because so few designs pass that there is nothing left to count, while rarefied U_{15} keeps climbing. The gap between those two curves is the whole argument for rarefaction. Every cell regenerates from `klsweep_table.py`, the single definition of these numbers.

What it shows. $w = 0.02$ buys $U_{15} = 803$ at 56.7 % pass. The dial is real and it is smooth over four orders of magnitude of weight.

What it can't say. One seed per arm. Two unseeded runs of one config on this benchmark gave 68.8 % and 89.6 % control pass — a 21-point swing, wider than the gap between the first two arms here.

3 C3 — The two dials are not on the same frontier

`C1 × C2 overlaid · same backbone, same reference, same` U_{15} `estimator`

C1 and C2 share a backbone, a reference sequence and a rarefaction depth, so their points go on one axis pair without rescaling. This is the comparison the paper is built on.

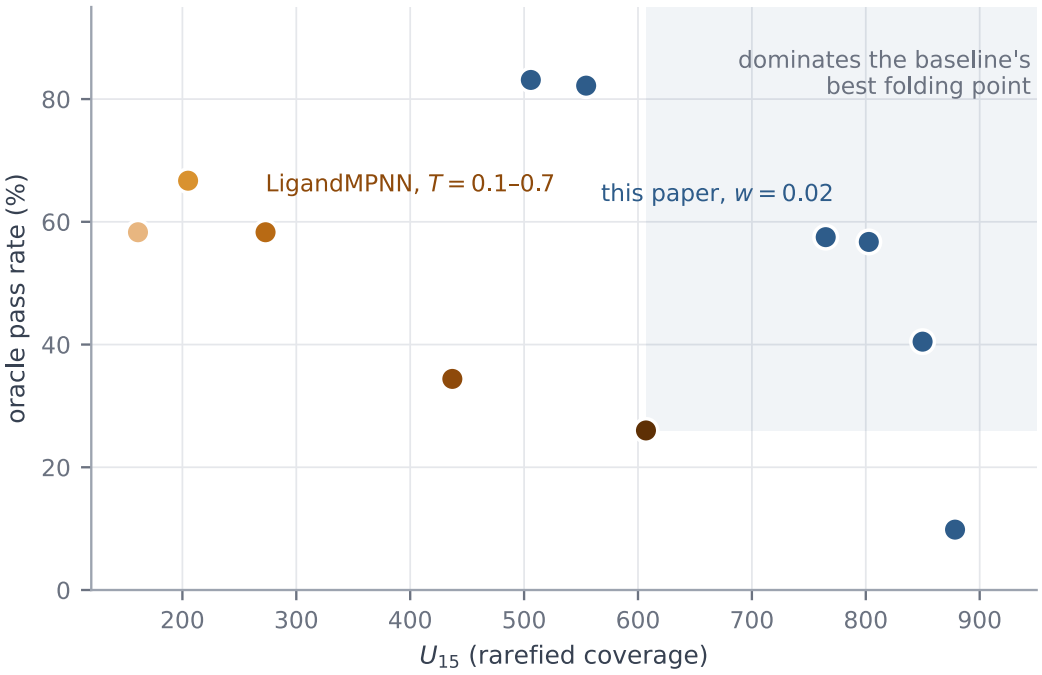


Figure 3: The shaded quadrant is everything that dominates the baseline's best folding point: $U_{15} > 607$ *and* pass $> 26.0\,\%$. Three KL arms sit inside it; no temperature setting does.

What it shows. A frontier *shift*, not a point win. RL does not merely pick a better spot on the existing trade — it moves the achievable set outward. At $w = 0.02$ the policy carries 1.32× the baseline's coverage at 2.2× its pass rate.

What it can't say. Mean pairwise Hamming at $w = 0.02$ is 114.4 against the baseline's 100.5 at $T = 0.7$: comparable, not better. The win is in *distinct substitutions at matched design count*, and the claim should be stated that narrowly.

4 C4 — Which part of the method did the work

`experiments/ame/..._conv200{_legacy, _surr, (none), _kl} · one backbone, four objectives`

An A/B ladder over the objective itself: the legacy reward, then the surrogate fix, the log-prob fix, and the KL anchor on top. Each row is the row above it plus one change.

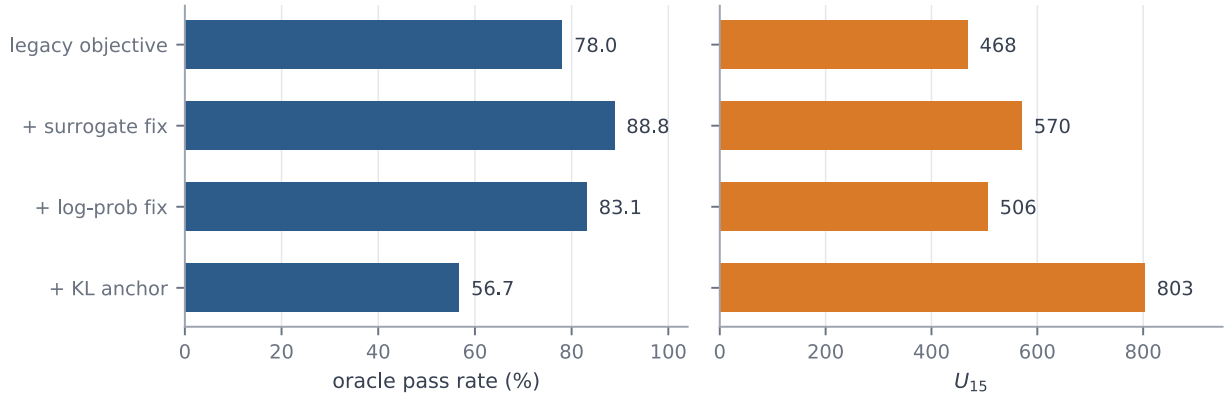


Figure 4: Two charts rather than two y-axes on one. The category order is the order the fixes were applied, which is itself information.

What it shows. The first three steps move pass rate around and barely touch coverage (468 → 571 → 506). **Only the KL anchor moves** U_{15}, by 297 points. The diversity result is attributable to one component.

What it can't say. The surrogate-fix arm is the *best* on pass rate (88.8 %) and is not the arm the paper headlines. All are single draws; the six replicates behind this family run 571, 510, 544 with the fix and 506, 511, 410 without — means 542 and 476, every one under the baseline's 607.

5 C5 — Recombining across backbones, easy to nearly impossible

`experiments/ame/r1panel/recomb · 9 backbones · 256 recombinants per cell, AF3 best-of-5`

The first campaign at library scale: cut passing designs into fragments, recombine across parents, fold the chimeras. Backbones are ordered by how dead the RFdiffusion2 benchmark finds them, which is the axis along which the result matters.

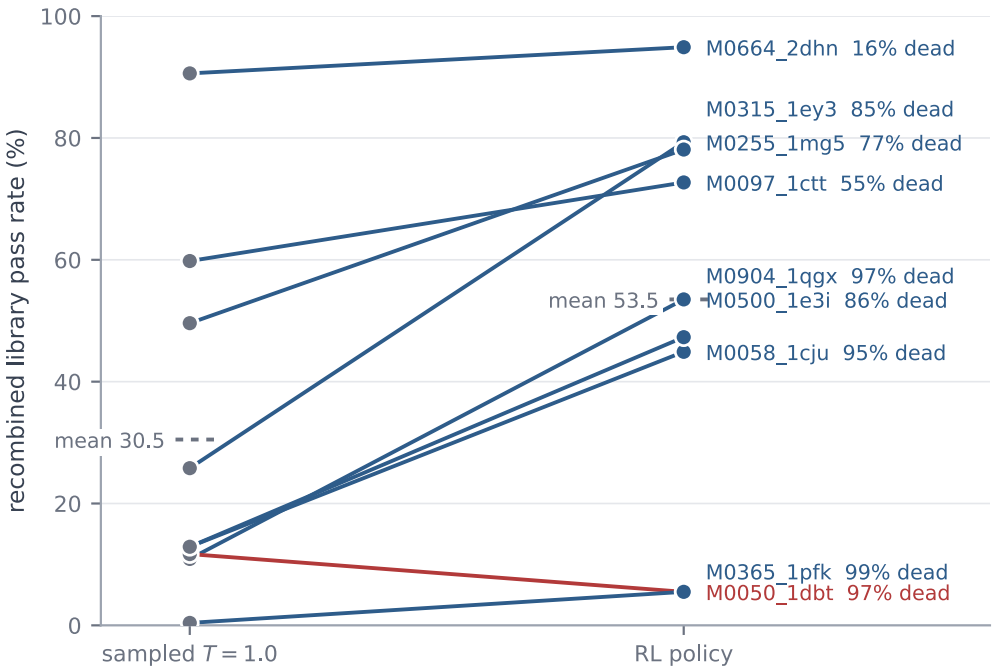


Figure 5: Each line is one backbone, sampled at $T = 1.0$ to the RL policy. Mean 53.5 % against 30.5 %, RL ahead on 8 of 9. The single loss, in red, is M0050_1dbt at 97 % dead, where both arms are in single digits.

What it shows. The gain is largest exactly where the benchmark is hardest — M0255_1mg5 at 77 % dead goes 25.8 to 79.3. An improvement on an easy backbone is a convenience; this is a different kind of claim.

What it can't say. Four backbones have no diversity reading — too few passing members on one arm to rarefy against — so the diversity comparison is 6 of 8, not 9 of 9. The two arms also differ in parent count, which is why C6 rarefies.

6 C6 — The combinatorial filter, as a Pareto

`r1panel + fragbsweep/fragb16 (k = 4) + fragk (k = 16) · 3 backbones · all arms batch 16`

Library pass rate against library diversity, for four ways of producing the parents. Both axes are properties of the *recombined* library. This is the panel the thesis stands on.

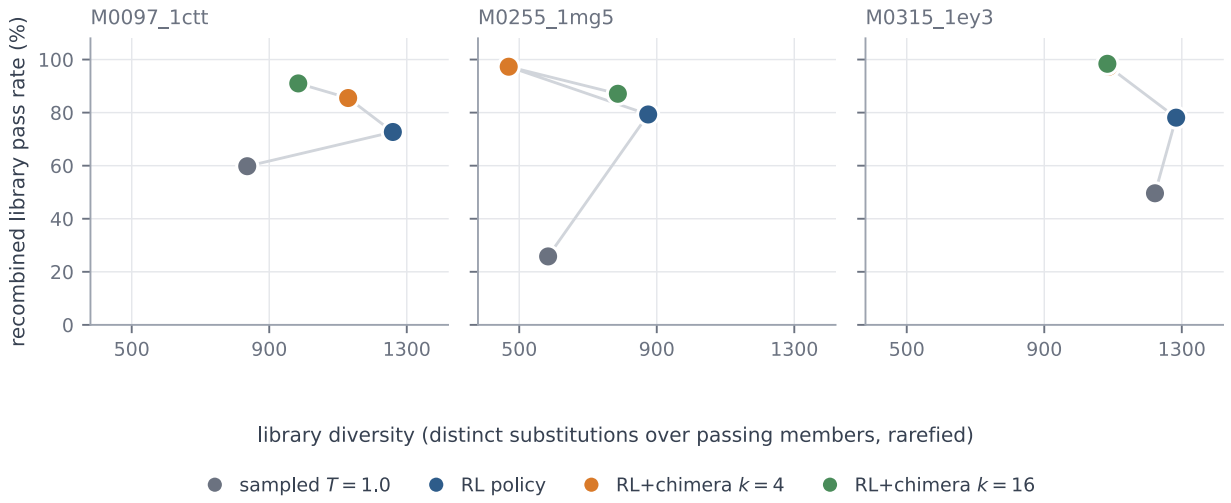


Figure 6: Faceted by backbone rather than overlaid: the arms cross each other in diversity, so one axis draws four long segments over one another and the shape of any single backbone becomes unreadable. Axes are shared. On M0315_1ey3 the two chimera arms very nearly coincide (1087.5/97.3 % and 1083.5/98.4 %) and plot as one point.

What it shows. RL moves up and right of sampled on all three backbones — a genuine dominance. The chimera objective then moves up again (93.4 % against 76.7 %, 3 of 3) **and back to the left**: diversity 895.5 against 1139.4, losing on 3 of 3. Cutting at $k = 16$ instead of 4 recovers part of it (951.6) at the same pass rate.

What it can't say. No arm here dominates every other. The chimera objective trades coverage for fidelity and k **is the dial along that trade**. The paper currently argues for the chimera arm on pass rate alone; this panel is why that has to be stated as a position on a frontier rather than as a win.

7 C7 — How much of C6 could be a lucky seed

`experiments/ame/{r1rep,fragb32} · T2 library eval · 256 parents + 256 recombinants per cell`

Two plain-RL replicates per backbone, differing only in `seed`, bound what a good draw buys. The chimera arm is then placed against that band.

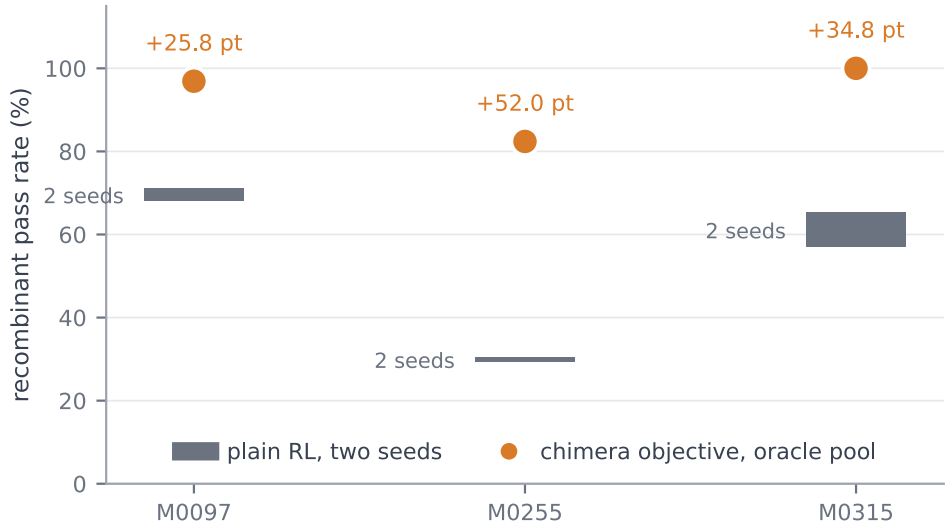


Figure 7: Bars span the two plain-RL seeds; the point is the chimera arm. Margins over the top of the band are +25.8, +52.0 and +34.8 points — three to six times the largest seed-to-seed difference.

What it shows. This is why a single chimera seed is worth something here when single-seed results usually are not: the effect clears the whole measured band on every backbone.

What it can't say. The band is measured on plain-RL/RF3 arms. **No pool arm has ever been run twice**, so the chimera point's own error bar is unmeasured. The contrast varies objective *and* oracle together and cannot apportion the gain. Quote the recombinant pass rate, never retention — retention swings 0.767 → 1.611 on an unchanged config.

8 C8 — Whether the training number was ever honest

`experiments/ame/{fragbsweep,fragb16,fragb32}/offline_probe · 38 held-out AF3 probe points`

The reward sits inside a structure predictor, so a training pass rate is not evidence on its own. Gap = oracle-balanced training level minus held-out AF3 on chimeras rebuilt deterministically from the same step's parents.

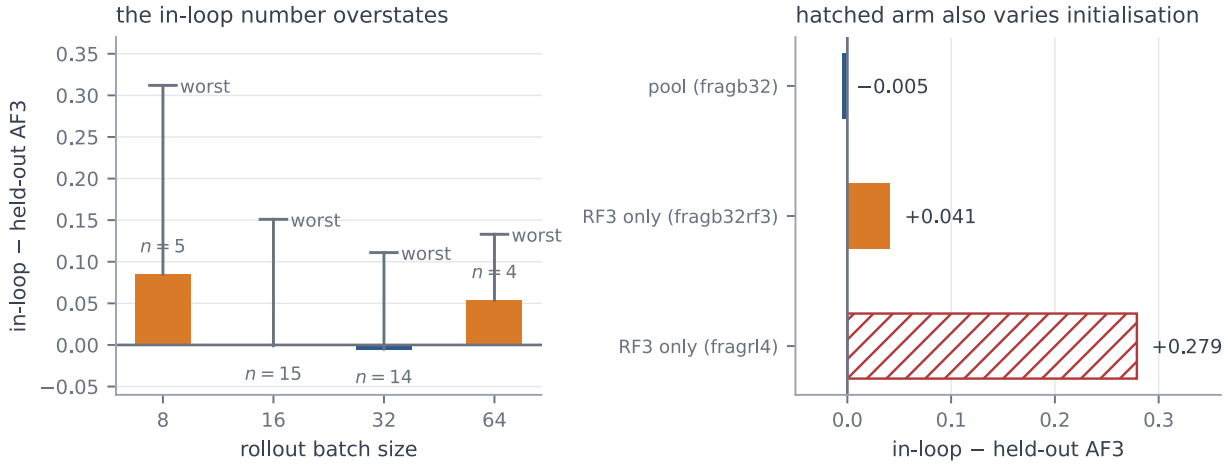


Figure 8: Above the zero line the training number overstates. Pooled over all 38 points: mean +0.015, median −0.000, with 7 of 38 overstating by more than 0.10. The pooled mean is carried by the two smallest cells, so it is quoted with its breakdown or not at all.

What it shows. At batch 16 and 32 the in-loop number is honest to within 0.005. The one contrast that isolates the oracle — `fragb32` against `fragb32rf3`, both batch 32, both chimera-objective, both self-resumed — puts **the oracle pool at** 0.046 of avoided overstatement.

What it can't say. The hatched arm, `fragrl4` at +0.279, warm-starts from a `fragrl3` checkpoint while the pool arms start fresh. It varies initialisation as well as oracle: it is the arm the project started from, never the controlled contrast. Neither arm in the controlled pair is replicated in seed.

9 C9 — The only campaign with a wet lab in it

`adaptyv_library · de novo PETase, 209 aa, 5 fragments · 3,135 wells assayed`

128 protein parts over five slots spanning 1.1×10^7 recombinants; 1,920 BsaI DNA blocks, one-pot assembly, one well per recombinant. 67 % **registered above background at a median of 29 mutations from the parent.**

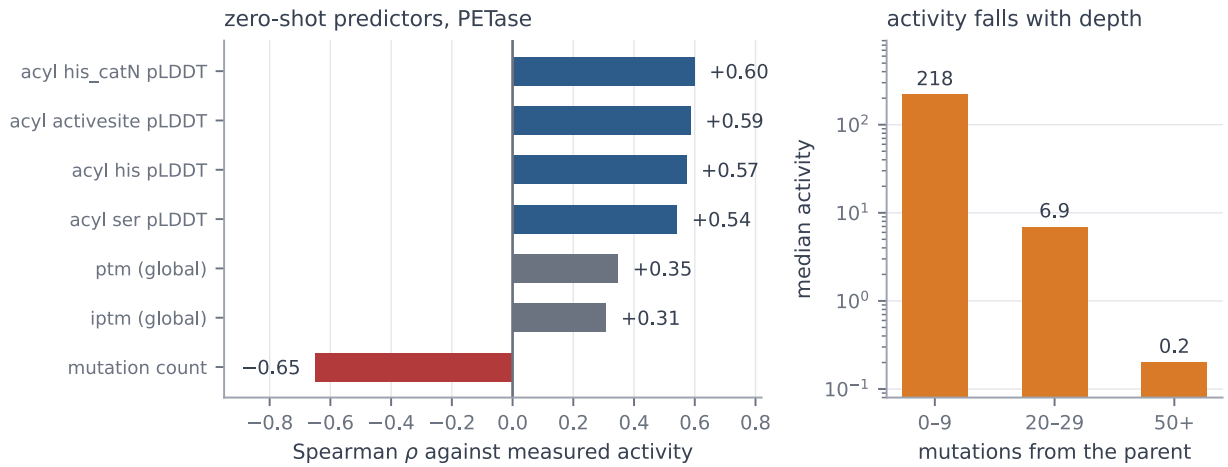


Figure 9: Left: scoring the right state at the right atoms is worth roughly 2× the global number. But distance from the parent ranks activity at −0.65, better than any AF3 metric, so the zero-shot claim is made only on the 2539 designs ≥ 20 mutations out, where active-site pLDDT still gives +0.59. Right: median activity by distance bin, log scale.

What it shows. The independence assumption that `results/fragments.tex` names as load-bearing survives a physical test. Searching the library does move activity: median 2.2 → 12.2 across four rounds, best recombinant 20.5 → 31.7, with the highest of all four (47.8) in round 2.

What it can't say. No wild-type comparator was assayed, so no fold-over-parent number exists — the archived 250× was always a placeholder. The 20 designs scoring above 100 are a different assay run with no shared normalising control, so nothing crosses that line. And on this scaffold recombination at depth bought reachable diversity and a 67 % register rate, *not* a better enzyme than local work.

10 C10 — The same pipeline on a second enzyme, and it inverts

`paper/figures/data/petase_super_*_transfer.csv · second de novo hydrolase, 160 aa, 2,509 purified`

Each point is one metric, plotted by how well it predicted activity on PETase against how well it predicted on **super**. On the diagonal the metric transfers; in the off-diagonal quadrants it points the wrong way.

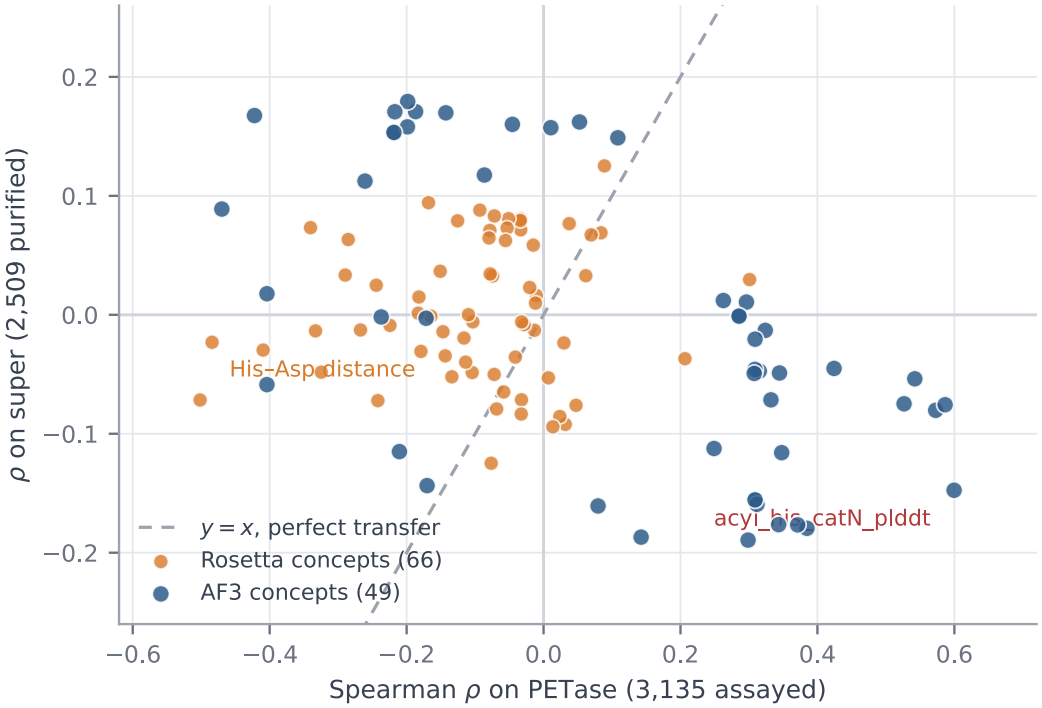


Figure 10: AF3 concepts: ρ-of-ρs $= -0.604$ over 49 shared concepts, with only 10 of 49 agreeing in sign against ∼24 expected by chance. All ten of PETase's strongest predictors invert, mean $+0.470 \rightarrow -0.100$. Rosetta fails more mildly — $+0.035$, 33 of 66 — i.e. no transferable signal rather than an inverted one.

What it shows. One family survives: PETase's three strongest Rosetta predictors are all His–Asp distance terms, and all three **keep their sign** while collapsing 7–21× in magnitude. Catalytic-geometry strain points the same way on both enzymes and nearly stops mattering — the only concept family that does.

What it can't say. This is the wet-lab counterpart of C8, and it is the strongest argument in the paper for the library strategy: a predictor's agreement with the quantity you care about is a property of the specific system, re-earned per target, never inherited. Because zero-shot scores do not transfer, selection has to be done by experiment — which is what a combinatorially compatible library at a 67 % register rate is for.

11 What a library costs

`design calculation, not a campaign · Twist eBlocks at $0.07 per base · 200 aa enzyme`

Every campaign above is eventually answerable to this one. A protein of L amino acids is $3L$ bases. Cut it into k equal fragments and each fragment is $3L/k$ bases, so a budget B at a price c per base buys

$$v = \frac{B}{3Lc} \quad \text{variants per slot,} \qquad S = v^k = \left(\frac{B}{3Lc}\right)^k \quad \text{recombinants.}$$

There is no k in v, and that is the whole point: cutting into k pieces buys k times as many fragments, each $1/k$ as long, and the two cancel exactly. k **is a pure exponent on a number set entirely by budget.** Doubling the budget multiplies the library by 2^k — at $k = 6$, twice the money is 64 times the library.

For a 200-aa enzyme at $0.07 per base, one full-length equivalent costs $42, so $10 000 stocks 238 variants in every slot regardless of how finely it is cut.

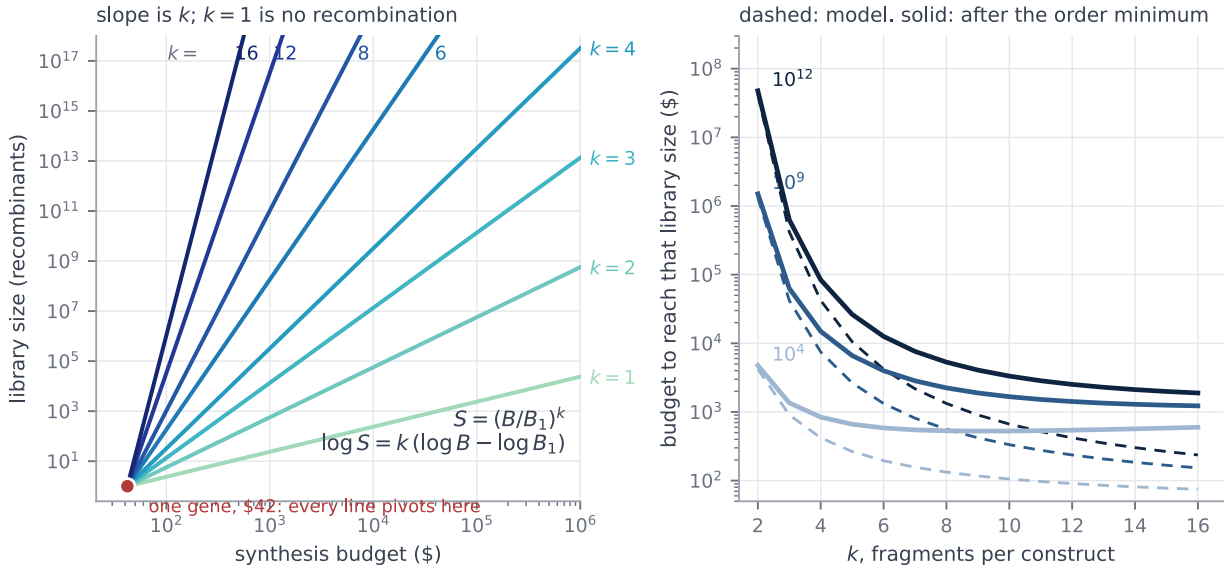


Figure 11: Left: taking logs, $\log S = k(\log B - \log B_1)$, so on log–log axes the entire model is a fan — every k is a straight line of slope k, and all of them pivot on the cost of one full-length gene. $k = 1$ is the line you get without recombination at all: buy v genes, have v sequences. The vertical distance from that line to any other is what cutting and reassembling buys for the same money. Because the axes are logarithmic a different price per base slides the pivot sideways and changes nothing else. Right: the same model inverted to the question actually asked — what a target library size costs, and how much of that is the vendor's order minimum rather than the model.

What it shows. The scaling is $S \propto B^k$, so k sets how steeply library size answers to money and nothing else in the model does. Going from no recombination to $k = 6$ turns a $10 000 order from 238 sequences into 1.8×10^{14}. Reaching 10^{12} recombinants costs $42 000 at $k = 4$ and $1329 at $k = 8$ — the same library, a factor of 32 cheaper, for cutting it twice as finely.

What it can't say. Nothing in this figure is measured; it is arithmetic on a price. It also says nothing about whether the library *works*, which is the entire subject of C6 and C7 — two libraries of identical size can differ by 50 points of pass rate. Read it as the budget axis of a decision whose other axes are elsewhere in this document.

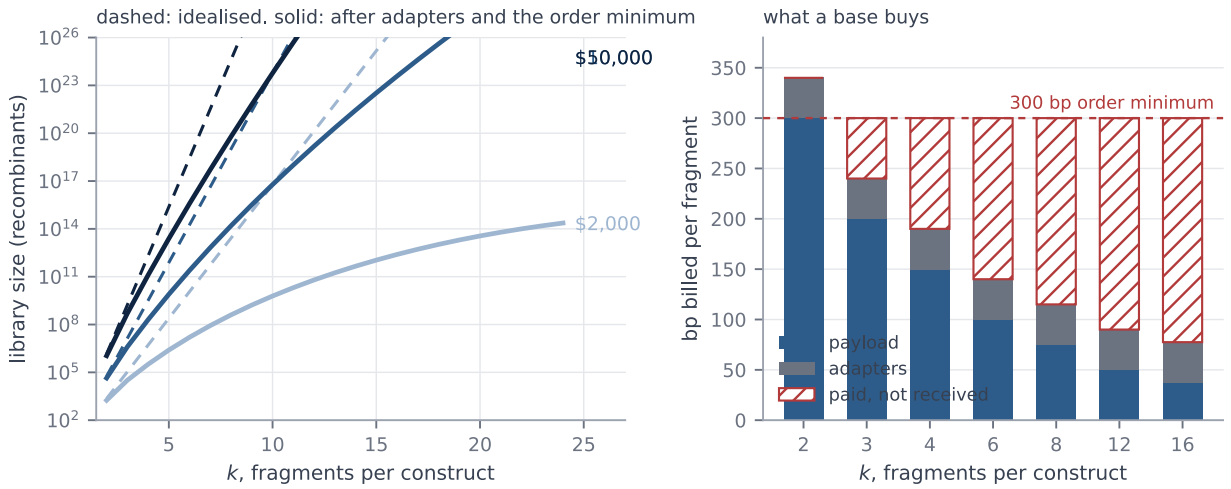


Figure 12: Left: the idealised exponent against what vendor terms actually deliver. Right: a fragment is billed at $\max(m,\, 3L/k + h)$, for an order minimum m and h bases of Golden Gate sites and overhangs. On a 200-aa enzyme the minimum binds from $k = 3$ onward, and by $k = 8$ three quarters of every base bought is padding. m **and** h **are assumed** — 300 and 40 bases — and are the two numbers to check against a live quote before anyone plans a spend from this figure. The $0.07 per base is not assumed.

What it shows. The order minimum is what converts k from free to expensive. Idealised, k costs nothing; after the minimum, going from $k = 4$ to $k = 16$ at $10 000 drops the variants per slot from 119 to 30, and the library still grows — 2×10^8 to 3.8×10^{23} — because the exponent beats the division. The penalty is real and it loses anyway.

What it can't say. So this calculation does *not* pick a k. Within everything synthesis charges for, larger k always wins, which means k has to be chosen on the axis C6 measures — where fidelity and coverage trade — and on how many junctions an assembly will actually tolerate. The cost model only says that money will not make that choice for you.

What binds instead. Three things run out before the budget does, and two of them are measured earlier in this document. **Parents:** stocking v variants in a slot needs v passing designs, so the 256-parent libraries of C6 and C7 are exhausted at $10 752 — past that point the policy is the limit, not the order. **Assembly:** $\sim 10^9$ transformants from a one-pot reaction. **Screening:** 3135 wells, in the one screen this project has run, against the 1.1×10^7 recombinants that library spanned. At $10 000 and $k = 6$ the model offers 1.8×10^{14}, which is five orders past what can be built and eleven past what can be read. Synthesis sets the ceiling of this programme; it has never set the floor.

12 Three arcs these panels will carry

Each arc names the panels that carry it and the panel that argues against it.

Arc 1 — RL moves the frontier. C1 → C2 → C3 as the spine, C5 for breadth, C6 for the library payoff. *Against it:* C4, where only one of four components moves coverage at all, and the best pass rate belongs to an arm the paper does not headline.

Arc 2 — fragments are the unit. C6 → C7 as the spine, C9 for physical realizability. *Against it:* C6 again — the chimera objective loses diversity on 3 of 3 backbones, so "recombinable" is bought, not free.

Arc 3 — you cannot predict, so you must order. C8 → C10 as the spine, with C6 as the thing you order and C9 as proof it assembles. *Against it:* nothing in these panels. It rests on the two strongest results in the project — 38 probe points and two independent enzyme screens — and it makes the diversity/fidelity frontier the deliverable rather than the claim.

What is not regenerable. Table 1's CLEO rows cannot be rebuilt: `backbone19` and `selection2` were pruned and no archive survives, which also breaks `ame_diversity.py`, `ame_frontier.py` and `ame_novelty.py`. The panels here avoid those trees entirely.